2 **(a)** State the principle of conservation of linear momentum.

……

…… [1]

(b) A light and long string (string 1), runs over two smooth and light pulleys (pulleys A and B). One end of the string is fixed to the ceiling and the other end is attached to mass M_1. Pulley A is fixed to the ceiling while pulley B is movable and is attached to mass M_2 via another light string (string 2).

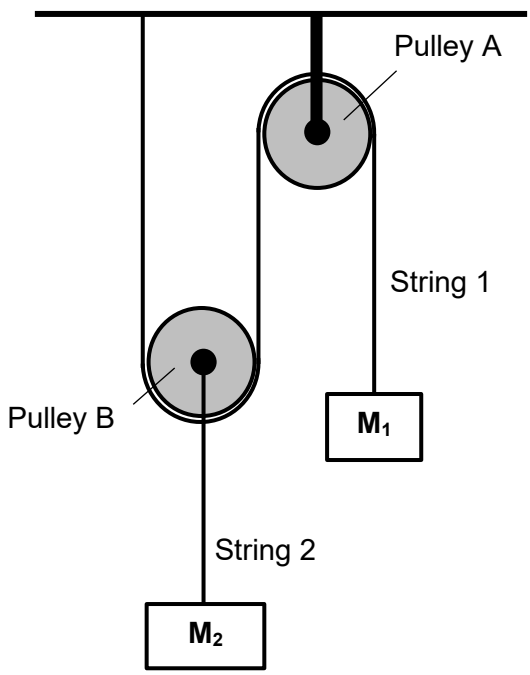

Fig. 2.1

(i) Given that M_1 = 4.0 kg and M_2 = 8.0 kg and they are both at rest, determine the tension in string 1 and string 2.

tension in string 1 = …… N

tension in string 2 = …… N [2]

(ii) A little disturbance is made to the system and M_1 starts to move upward with speed v_1 while M_2 starts to move downwards with v_2. By considering linear momentum, express v_2 in terms of v_1.

v_2 = ... [2]

(iii) Hence, determine the energy introduced to the system due to the disturbance in terms of v_1.

energy = ... [2]

(iv) After some time, the masses move with constant speed. Show that total energy of the system is conserved. Explain your working clearly. [2]